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Master in Computer Applications

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Class: II Sem

Course: Database Management System

Team : 01

Assignment-1

Conceptual Design of Database for GMU Event Management System

Assignment Title : Designing Conceptual Schema for GMU Event Management System.

Introduction:

GM University hosts multiple events throughout the year, ranging from academic seminars to cultural festivals. The goal of this assignment is to lay the groundwork by understanding DBMS concepts and modeling the requirements using a conceptual data model.

Problem Statement:

The university lacks a centralized system to store and manage data related to its events, such as coordinators, attendees, event schedules, and resources. Without a proper database structure, event planning and management become inefficient.

Problem Solving:

- Explore the fundamentals of DBMS, types of data models, DBMS architecture, and relational schema.
- Identify the core entities: Event, Organizer, Venue, Participant, Feedback, etc.
- Use the Entity-Relationship (ER) model to define relationships, keys, and attributes.
- Identify primary and foreign keys to establish data integrity.

Presentation of Solution:

- Create an ER diagram for the event management system.
- Provide relational schemas derived from the ER model.
- Tabulate the list of entities, their attributes, and relationships.
- Discuss DBMS concepts applied in modeling.

Results and Discussions:

Discuss how the ER model represents real-world entities and how this approach supports efficient database design. Highlight limitations or assumptions made.

Applications:

- Used in designing university event registration systems.
- This conceptual model will serve as the blueprint for developing the physical database and implementing SQL queries in the next phase.

Introduction :

GM University (GMU) organizes diverse events across academic, cultural, and professional domains. These include seminars, symposiums, cultural festivals, intercollegiate competitions, alumni meets, webinars, and more. Each event involves multiple stakeholders such as organizers, participants, venue managers, and feedback handlers. However, the university currently lacks a centralized digital system for organizing, storing, and managing this critical event data.

The purpose of this assignment is to design a conceptual schema using Database Management System (DBMS) principles. This schema acts as a foundational blueprint for future implementation phases, ensuring better efficiency, data integrity, and scalability in managing GMU's event-related operations.

Problem Statement:

The university lacks a centralized system to store and manage data related to its events, such as coordinators, attendees, event schedules, and resources. Without a proper database structure, event planning and management become inefficient.

Problem Solving :**1. Explore DBMS Concepts**

- DBMS Architecture: We use the three-schema architecture:
 - External Schema (User views: Admin, Organizer, Participant)
 - Conceptual Schema (Design level: Entity-Relationship Model)
 - Internal Schema (Physical implementation – planned in later stages)
- Types of Data Models:
 - Hierarchical, Network, Relational — *We use the Relational Model for its simplicity, scalability, and widespread use.*
- ER Model: Utilized to map real-world entities (Event, Organizer, etc.) into structured elements such as attributes, relationships, and constraints.

2. Identify Core Entities and Attributes

- Event: EventID, Name, Date, Type, Description, VenueID, OrganizerID
- Organizer: OrganizerID, Name, Email, Contact
- Participant: ParticipantID, Name, Email, Phone
- Venue: VenueID, Name, Location, Capacity
- Registration: Composite relationship (EventID, ParticipantID), Date
- Feedback: FeedbackID, EventID, ParticipantID, Rating, Comments
- Resource: ResourceID, Name, Quantity, AssignedEventID

3. Define Relationships

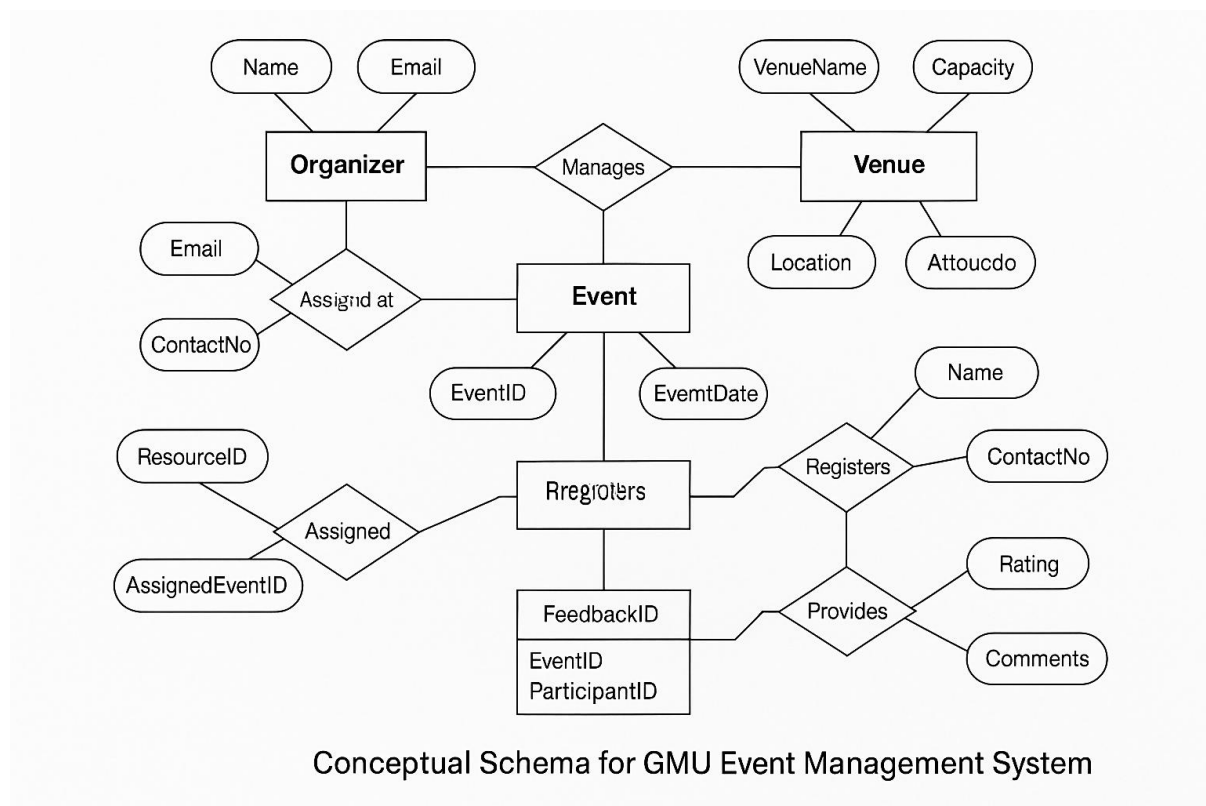
- Each Event is managed by an Organizer.
- Events are hosted at a Venue.
- Participants register for multiple Events.
- Each Registration can submit Feedback for the Event.
- Events require Resources, which are allocated as needed.

4. Keys for Data Integrity

- Primary Keys (PK): Unique identifiers for each entity (e.g., EventID)
- Foreign Keys (FK): Maintain referential integrity (e.g., VenueID in Event refers to Venue)

Presentation of Solution :

1. ER Diagram :



2. Relational Schemas

Entity/Relationship	Attributes	Primary Key (PK)	Foreign Keys (FK)
Event	EventID, EventName, EventDate, EventType, Description, OrganizerID, VenueID	EventID	OrganizerID → Organizer, VenueID → Venue
Organizer	OrganizerID, Name, Email, ContactNo	OrganizerID	-
Participant	ParticipantID, Name, Email, ContactNo	ParticipantID	-

Entity/Relationship	Attributes	Primary Key (PK)	Foreign Keys (FK)
Venue	VenueID, VenueName, Location, Capacity	VenueID	-
Registration	EventID, ParticipantID, RegistrationDate	(EventID, ParticipantID)	EventID → Event, ParticipantID → Participant
Feedback	FeedbackID, EventID, ParticipantID, Rating, Comments	FeedbackID	EventID → Event, ParticipantID → Participant
Resource	ResourceID, ResourceName, Quantity, AssignedEventID	ResourceID	AssignedEventID → Event

3. Tabulated Entities and Relationships :

Entity	Relationships	Cardinality	Explanation
Event	Organizer manages Event	1:N	One organizer manages multiple events
Event	Venue hosts Event	1:N	One venue can host multiple events (sequentially)
Participant	Registers for Event	M:N	Participants can join multiple events
Registration	Linked to Feedback	1:1	Each registration can result in one feedback
Resource	Assigned to Event	N:1	Resources are allocated per event

Attributes And Entities :

Organizer :

```

2 • CREATE TABLE Organizer (
3     OrganizerID INT PRIMARY KEY,
4     Name VARCHAR(100),
5     Email VARCHAR(100),
6     ContactNo VARCHAR(15)
7 );
8 • desc Organizer;
9

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Field	Type	Null	Key	Default	Extra
►	OrganizerID	int	NO	PRI	NULL	
	Name	varchar(100)	YES		NULL	
	Email	varchar(100)	YES		NULL	
	ContactNo	varchar(15)	YES		NULL	

Result Grid | Filter Rows: | Edit:   

	OrganizerID	Name	Email	ContactNo
►	1	Dr. Mehta	mehta@gmu.edu	9876543210
	2	Prof. R. Sharma	sharma@gmu.edu	9123456780
	3	Anjali Rao	anjali.rao@gmu.edu	9988776655
	4	Manoj Desai	manoj@gmu.edu	8877665544
	5	Ritika Nair	ritika@gmu.edu	9090909090
	6	Vikram Joshi	vikram@gmu.edu	9898989898
	7	Neha Kulkarni	neha@gmu.edu	9123456789
	8	Amit Verma	amit@gmu.edu	8765432109
	9	Pooja Iyer	pooja@gmu.edu	9234567890
	10	Suresh Patil	suresh@gmu.edu	9345678901
*	NULL	NULL	NULL	NULL

Venue :

```

10 • ○ CREATE TABLE Venue (
11       VenueID INT PRIMARY KEY,
12       VenueName VARCHAR(100),
13       Capacity INT,
14       Location VARCHAR(100)
15     );
16 • desc Venue;

```

Result Grid Filter Rows: Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	VenueID	int	NO	PRI	NULL	
	VenueName	varchar(100)	YES		NULL	
	Capacity	int	YES		NULL	
	Location	varchar(100)	YES		NULL	

	VenueID	VenueName	Capacity	Location
▶	1	Main Auditorium	500	Block A
	2	Conference Hall	200	Block B
	3	Outdoor Ground	1000	Backyard
	4	Library Seminar Room	100	Block C
	5	Cultural Hall	300	Block D
	6	Computer Lab	50	Block E
	7	Exhibition Hall	400	Block F
	8	Sports Complex	800	Sports Block
	9	Online Webinar Room	1000	Virtual
	10	Lecture Hall 5	150	Block G
•	NULL	NULL	NULL	NULL

Event :

```

17 • CREATE TABLE Event (
18     EventID INT PRIMARY KEY,
19     EventName VARCHAR(100), EventDate DATE,
20     EventType VARCHAR(50), Description TEXT, OrganizerID INT, VenueID INT,
21     FOREIGN KEY (OrganizerID) REFERENCES Organizer(OrganizerID),
22     FOREIGN KEY (VenueID) REFERENCES Venue(VenueID)
23 );
24 • desc Event;

```

Field	Type	Null	Key	Default	Extra
EventID	int	NO	PRI	NULL	
EventName	varchar(100)	YES		NULL	
EventDate	date	YES		NULL	
EventType	varchar(50)	YES		NULL	
Description	text	YES		NULL	
OrganizerID	int	YES	MUL	NULL	
VenueID	int	YES	MUL	NULL	

EventID	EventName	EventDate	EventType	Description	OrganizerID	VenueID
1	Tech Talk	2025-08-01	Seminar	AI in Education	1	1
2	Cultural Fest	2025-08-10	Festival	Annual cultural event	2	5
3	Hackathon	2025-08-20	Competition	24-hour coding challenge	3	2
4	Book Exhibition	2025-08-25	Exhibition	Latest books and publications	4	7
5	Sports Meet	2025-09-01	Sports	Inter-department sports	5	8
6	Webinar on Cybersecurity	2025-09-05	Webinar	Cybersecurity awareness	6	9
7	Alumni Meet	2025-09-10	Meet	Reunion for alumni	7	3
8	Photography Workshop	2025-09-15	Workshop	Basics of photography	8	6
9	Career Fair	2025-09-20	Fair	Opportunities from various companies	9	10
10	Math Quiz	2025-09-25	Quiz	Quiz competition on mathematics	10	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Participant :

```

25 • CREATE TABLE Participant (
26     ParticipantID INT PRIMARY KEY,
27     Name VARCHAR(100),
28     Email VARCHAR(100),
29     ContactNo VARCHAR(15)
30 );
31 • desc Participant;
32

```

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content:
	Field	Type	Null	Key	Default	Extra
▶	ParticipantID	int	NO	PRI	NULL	
	Name	varchar(100)	YES		NULL	
	Email	varchar(100)	YES		NULL	
	ContactNo	varchar(15)	YES		NULL	

	ParticipantID	Name	Email	ContactNo
▶	1	Aarav Gupta	aarav@gmu.edu	9876501234
	2	Isha Rao	isha@gmu.edu	9876505678
	3	Karan Mehta	karan@gmu.edu	9876506789
	4	Priti Deshmukh	priti@gmu.edu	9876507890
	5	Yash Patel	yash@gmu.edu	9876508901
	6	Meera Singh	meera@gmu.edu	9876512345
	7	Raj Khanna	raj@gmu.edu	9876513456
	8	Tanvi Joshi	tanvi@gmu.edu	9876514567
	9	Rohan Verma	rohan@gmu.edu	9876515678
	10	Sneha Nair	sneha@gmu.edu	9876516789
•	NULL	NULL	NULL	NULL

Registration :

The screenshot shows a database management tool interface. The top part displays SQL code for creating and describing the 'Registration' table. The bottom part shows a 'Result Grid' with columns: Field, Type, Null, Key, Default, and Extra.

```

33 • CREATE TABLE Registration (
34     RegistrationID INT PRIMARY KEY,
35     EventID INT, ParticipantID INT, RegistrationDate DATE,
36     FOREIGN KEY (EventID) REFERENCES Event(EventID),
37     FOREIGN KEY (ParticipantID) REFERENCES Participant(ParticipantID)
38 );
39 • desc Registration;
40

```

Field	Type	Null	Key	Default	Extra
RegistrationID	int	NO	PRI	NULL	
EventID	int	YES	MUL	NULL	
ParticipantID	int	YES	MUL	NULL	
RegistrationDate	date	YES		NULL	

	RegistrationID	EventID	ParticipantID	RegistrationDate
▶	1	1	1	2025-07-20
	2	1	2	2025-07-21
	3	2	3	2025-07-22
	4	3	4	2025-07-23
	5	4	5	2025-07-24
	6	5	6	2025-07-25
	7	6	7	2025-07-26
	8	7	8	2025-07-27
	9	8	9	2025-07-28
	10	9	10	2025-07-29
✱	NULL	NULL	NULL	NULL

Feedback :

```

41 • CREATE TABLE Feedback (
42     FeedbackID INT PRIMARY KEY, EventID INT, ParticipantID INT,
43     Rating INT CHECK (Rating BETWEEN 1 AND 5), Comments TEXT,
44     FOREIGN KEY (EventID) REFERENCES Event(EventID),
45     FOREIGN KEY (ParticipantID) REFERENCES Participant(ParticipantID)
46 );
47 • desc Feedback;
48

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Field	Type	Null	Key	Default	Extra
►	FeedbackID	int	NO	PRI	NULL	
	EventID	int	YES	MUL	NULL	
	ParticipantID	int	YES	MUL	NULL	
	Rating	int	YES		NULL	
	Comments	text	YES		NULL	

	FeedbackID	EventID	ParticipantID	Rating	Comments
►	1	1	1	5	Very informative seminar
	2	1	2	4	Good presentation
	3	2	3	5	Enjoyed the event
	4	3	4	3	Could be better
	5	4	5	4	Loved the variety of books
	6	5	6	5	Great arrangements
	7	6	7	4	Very useful content
	8	7	8	5	Wonderful interaction with alumni
	9	8	9	3	Too technical
	10	9	10	4	Nice opportunities
•	NULL	NULL	NULL	NULL	NULL

Resource :

```

49 • CREATE TABLE Resource (
50     ResourceID INT PRIMARY KEY,
51     ResourceName VARCHAR(100),
52     Quantity INT,
53     AssignedEventID INT,
54     FOREIGN KEY (AssignedEventID) REFERENCES Event(EventID)
55 );
56 • desc Resource;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	ResourceID	int	NO	PRI	NULL	
	ResourceName	varchar(100)	YES		NULL	
	Quantity	int	YES		NULL	
	AssignedEventID	int	YES	MUL	NULL	

	ResourceID	ResourceName	Quantity	AssignedEventID
▶	1	Projector	2	1
	2	Speakers	4	1
	3	Chairs	300	2
	4	Tents	10	3
	5	Tables	20	4
	6	Sports Equipment	50	5
	7	Mic Systems	3	6
	8	Laptops	10	3
	9	Cameras	5	8
	10	Banner Stands	15	9
*	NULL	NULL	NULL	NULL

Results and Discussions

➤ ER Model Validates Real-World Operations

The ER model precisely maps the real-world event management process at GMU by:

- Identifying distinct roles (organizer, participant) and objects (event, venue, resource).
- Structuring many-to-many (M:N) relationships like participant registrations through the use of a junction table (Registration).
- Capturing feedback and engagement through the Feedback entity, which connects back to specific events and participants.
- Enabling linkage between resources and events for logistical support.
- ✓ This shows that the model is logically sound and mimics real organizational behavior.

➤ Data Integrity Maintained Through Constraints

The database ensures referential integrity through:

- Primary keys (PK) to uniquely identify each record.
- Foreign keys (FK) to prevent invalid entries (e.g., no orphan records in Event, Registration, or Feedback).
- CHECK constraints to ensure valid feedback ratings (between 1 and 5).
- ✓ This reduces errors, prevents duplication, and ensures consistent and clean data entry.

➤ Scalability and Extensibility

The current model is scalable for:

- Adding multiple departments or clubs as part of the Organizer entity.
- Including sub-events, sponsors, or event categories as future entities.
- Integrating with web or mobile platforms for live registrations and feedback.
- ✓ The normalized structure allows for easy modifications without major schema redesign.

➤ Efficient Querying and Reporting

The relational schema supports:

- Event-wise participant listings.
- Event popularity analysis using feedback ratings.
- Resource utilization tracking for logistics planning.
- Monthly reports for event frequency, attendance, and feedback trends.
- ✓ This aids administrators in strategic planning and resource optimization

➤ **Minimizes Redundancy and Anomalies**

The design follows 3rd Normal Form (3NF) principles:

- No repeating groups (1NF satisfied)
- Non-key attributes depend only on primary keys (2NF)
- No transitive dependencies (3NF)
- ✓ This leads to compact, consistent, and update-efficient tables.

➤ **Improved Collaboration and Coordination**

By creating a centralized system:

- Organizers can view available venues and avoid conflicts.
- Participants can register easily and receive real-time updates.
- Administrators can assign resources logically and prevent overbooking.
- ✓ This fosters better coordination among departments, leading to smoother event execution.

➤ **Real-World Implementation Potential**

This database model is implementation-ready using platforms like:

- MySQL/PostgreSQL for backend development.
- Django/Flask or PHP for web portal integration.
- Power BI/Tableau for analytics and dashboards.
- ✓ The conceptual schema is aligned with industry practices and can be deployed in educational institutions or corporate environments.

➤ **Performance & Optimization (Future Scope)**

Though conceptual, performance can be enhanced in later stages by:

- Indexing on frequently searched fields (e.g., EventDate, EventType).
- Adding stored procedures for reporting.
- Including triggers for automated notifications (e.g., feedback reminders).
- ✓ Design choices in this model support long-term performance and maintainability.

Applications :

- **University Event Portals:** Easily adapted for student councils, alumni cells, placement teams.
- **Automated Registrations:** Simplifies user enrollment and confirmation workflows.
- **Resource Optimization:** Tracks how university assets (projectors, halls) are used during events.
- **Feedback Analytics:** Helps in evaluating event quality and participant satisfaction.
- **Foundation for SQL Queries & Reports:** This model sets the stage for database normalization, query writing, and interface development.

Conclusion :

The design of the Conceptual Schema for the GMU Event Management System demonstrates how database principles can solve real-world institutional problems. Through the use of ER modeling and relational schemas, the university can lay the groundwork for a powerful digital system that transforms event planning, participation, and analysis. This model ensures scalability, reduces redundancy, and supports future enhancements such as real-time updates, role-based access, and automated notifications.